

A - Second Half Sum

Time Limit: 2 sec / Memory Limit: 1024 MiB

Score : 100 points

Problem Statement

You are given an integer sequence of length N : $A = (A_1, A_2, \dots, A_N)$. Here, N is even.
Find the sum of the latter half of A , that is, the sum of $A_{(N/2)+1}, A_{(N/2)+2}, \dots, A_N$.

Constraints

- All input values are integers.
- N is an even number satisfying $2 \leq N \leq 100$.
- $1 \leq A_i \leq 100$

Input

The input is given from Standard Input in the following format:

```
N
A_1 A_2 ... A_N
```

Output

Output the answer.

Sample Input 1

```
8
1 3 7 8 4 2 5 6
```

Sample Output 1

```
17
```

The length of the sequence A is $N = 8$, and you should find the sum of A_5, A_6, A_7, A_8 .
Since $A_5 + A_6 + A_7 + A_8 = 4 + 2 + 5 + 6 = 17$, output 17.

Sample Input 2

```
2
1 100
```

Sample Output 2

```
100
```

Sample Input 3

```
10
31 41 59 26 53 58 97 93 23 84
```

Sample Output 3

```
355
```

B - Old Maid

Time Limit: 2 sec / Memory Limit: 1024 MiB

Score : 200 points

Problem Statement

Takahashi currently has N cards. The i -th ($1 \leq i \leq N$) card has the integer A_i written on it.

He repeats the following operation as many times as possible.

- Choose two different cards with the same integer written on them, and eat those two cards. The eaten cards are permanently removed and cannot be chosen in subsequent operations.

Find the sum of the integers written on the cards that remain when the operation can no longer be performed.

Constraints

- $1 \leq N \leq 100$
- $1 \leq A_i \leq 100$ ($1 \leq i \leq N$)
- All input values are integers.

Input

The input is given from Standard Input in the following format:

```
 $N$   
 $A_1$   $A_2$   $\cdots$   $A_N$ 
```

Output

Output the sum of the integers written on the remaining cards.

Sample Input 1

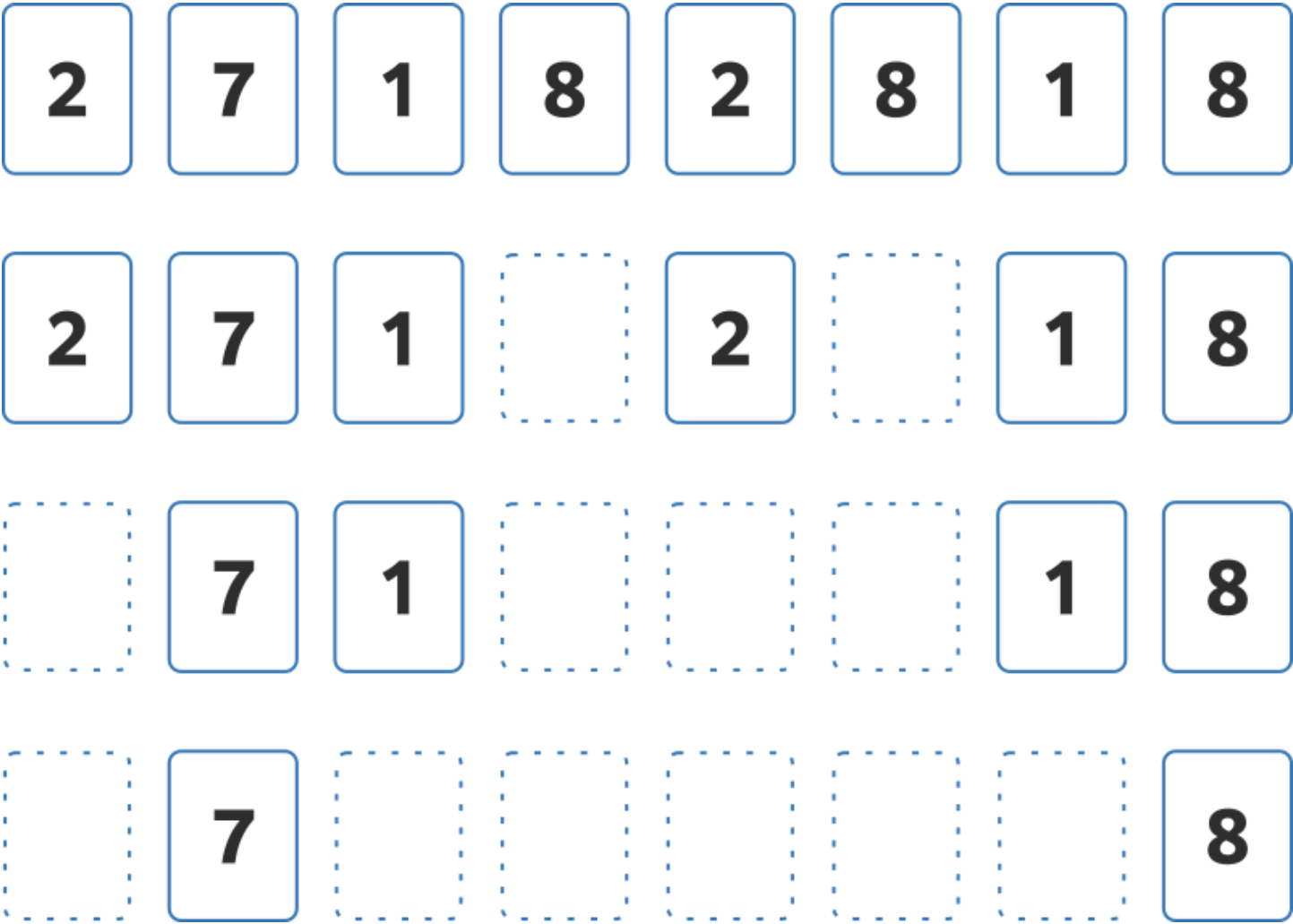
```
8  
2 7 1 8 2 8 1 8
```

Sample Output 1

15

For example, Takahashi operates as follows.

- 1. The fourth card and sixth card both have 8 written on them, so he chooses them and eats them.
- 2. The first card and fifth card both have 2 written on them, so he chooses them and eats them.
- 3. The third card and seventh card both have 1 written on them, so he chooses them and eats them.



Thus, the sum of the integers written on the remaining cards is $7 + 8 = 15$.

Note that there are three cards with 8 written on them, so once two of them are eaten, the remaining one cannot be eaten.

Sample Input 2

5
1 2 3 4 5

Sample Output 2

```
15
```

It is possible that Takahashi cannot perform the operation even once.

Sample Input 3

```
15
58 97 74 16 97 74 97 16 51 52 58 52 74 32 43
```

Sample Output 3

```
297
```

C - Change Schools

Time Limit: 2 sec / Memory Limit: 1024 MiB

Score : 300 points

Problem Statement

Currently, AtCoder High School has K classes and N students, and the i -th ($1 \leq i \leq N$) student belongs to the A_i -th class.

Takahashi is transferring to AtCoder High School in September. At that time, he can choose any one class among the K classes and belong to that class.

He will be sad if there is a class with more students belonging to it than the class he belongs to. Otherwise, he will be happy.

Find how many classes make him happy if he belongs to them.

Constraints

- $1 \leq N \leq 2 \times 10^5$
- $1 \leq K \leq N$
- $1 \leq A_i \leq K$ ($1 \leq i \leq N$)
- All input values are integers.

Input

The input is given from Standard Input in the following format:

```
N K  
A_1 A_2 ... A_N
```

Output

Output the number of classes satisfying the condition.

Sample Input 1

```
8 5  
3 3 5 5 4 4 3 2
```

Sample Output 1

```
3
```

For example, if Takahashi chooses the fifth class and belongs to it, the number of students belonging to the fifth class becomes 3. Then, he will be happy since there is no class with more than three students belonging to it.

On the other hand, for example, if he chooses the first class and belongs to it, the number of students belonging to the first class becomes 1. Then, he will be sad since three students belong to the third class.

He will be happy if and only if he chooses the third, fourth, or fifth class, so output 3.

Sample Input 2

```
6 1
1 1 1 1 1 1
```

Sample Output 2

```
1
```

It is possible that AtCoder High School has only one class.

Sample Input 3

```
14 8
6 1 5 3 8 4 3 4 3 5 1 2 5 1
```

Sample Output 3

```
4
```

D - Coefficient Stair

Time Limit: 2 sec / Memory Limit: 1024 MiB

Score : 400 points

Problem Statement

Output all length- N sequences $A = (A_1, A_2, \dots, A_N)$ consisting of non-negative integers that satisfy $\sum_{i=1}^N i \times A_i = K$, in lexicographic order from smallest to largest.

Here, you will only receive inputs such that the number of sequences satisfying the condition is at most 3×10^5 .

► What is lexicographic order for sequences?

Constraints

- $1 \leq N \leq 10$
- $1 \leq K \leq 2 \times 10^5$
- The number of sequences satisfying the condition is at most 3×10^5 .
- All input values are integers.

Input

The input is given from Standard Input in the following format:

N K

Output

Let q be the number of sequences of non-negative integers satisfying the condition; output them over q lines. Each line should contain the elements of a sequence of non-negative integers satisfying the condition, in order, separated by spaces. For every sequence, all sequences outputted before it must be lexicographically smaller than it.

Sample Input 1

3 8

Sample Output 1

```
0 1 2
0 4 0
1 2 1
2 0 2
2 3 0
3 1 1
4 2 0
5 0 1
6 1 0
8 0 0
```

For example, for the sequence $(0, 1, 2)$ we have $0 \times 1 + 1 \times 2 + 2 \times 3 = 0 + 2 + 6 = 8$, so it satisfies the condition. There is no sequence satisfying the condition that is lexicographically smaller than this one, so output `0 1 2` on the first line.

Including $(0, 1, 2)$, ten sequences satisfy the condition. Output these in lexicographic order from smallest to largest.

Sample Input 2

```
1 200000
```

Sample Output 2

```
200000
```

Sample Input 3

```
8 9
```

Sample Output 3

```
0 0 0 1 1 0 0 0
0 0 1 0 0 1 0 0
0 0 3 0 0 0 0 0
0 1 0 0 0 0 1 0
0 1 1 1 0 0 0 0
0 2 0 0 1 0 0 0
0 3 1 0 0 0 0 0
1 0 0 0 0 0 0 1
1 0 0 2 0 0 0 0
1 0 1 0 1 0 0 0
1 1 0 0 0 1 0 0
1 1 2 0 0 0 0 0
1 2 0 1 0 0 0 0
1 4 0 0 0 0 0 0
2 0 0 0 0 0 1 0
2 0 1 1 0 0 0 0
2 1 0 0 1 0 0 0
2 2 1 0 0 0 0 0
3 0 0 0 0 1 0 0
3 0 2 0 0 0 0 0
3 1 0 1 0 0 0 0
3 3 0 0 0 0 0 0
4 0 0 0 1 0 0 0
4 1 1 0 0 0 0 0
5 0 0 1 0 0 0 0
5 2 0 0 0 0 0 0
6 0 1 0 0 0 0 0
7 1 0 0 0 0 0 0
9 0 0 0 0 0 0 0
```

E - K-Divisible Subarrays

Time Limit: 2 sec / Memory Limit: 1024 MiB

Score : 450 points

Problem Statement

You are given a sequence of length N consisting of non-negative integers: $A = (A_1, A_2, \dots, A_N)$.

Define the **score** of a sequence of length 1 or more consisting of sequences of non-negative integers, $S = (S_1, S_2, \dots, S_k)$, as the number of sequences S_i ($1 \leq i \leq k$) whose sum of elements is divisible by K .

Find the maximum possible score of a sequence of non-negative integer sequences obtained by dividing A into one or more contiguous subsequences. Here, dividing a sequence A of length N into one or more contiguous subsequences means choosing a sequence of integers $l = (l_1, l_2, \dots, l_k)$ ($1 = l_1 < l_2 < \dots < l_k \leq N$) of length 1 or more, and forming the following k sequences.

- $(A_{l_i}, A_{l_i+1}, \dots, A_{l_{i+1}-1})$ ($1 \leq i \leq k$) (Here, let $l_{k+1} = N + 1$.)

Constraints

- $1 \leq N \leq 2 \times 10^5$
- $1 \leq K \leq 10^9$
- $0 \leq A_i < K$
- All input values are integers.

Input

The input is given from Standard Input in the following format:

```
N K
A_1 A_2 ... A_N
```

Output

Output the answer.

Sample Input 1

```
6 10
6 8 2 2 6 4
```

Sample Output 1

2

For example, A can be divided into four contiguous subsequences as $(6), (8, 2), (2), (6, 4)$. The sums of the elements of the second and fourth sequences are multiples of 10, so the score of $((6), (8, 2), (2), (6, 4))$ is 2.

A cannot be divided so that the score is 3 or more, so output 2.

Sample Input 2

8 1
0 0 0 0 0 0 0 0

Sample Output 2

8

Sample Input 3

30 8
5 0 4 2 7 3 2 3 2 4 0 1 4 0 4 1 7 5 2 5 0 3 6 6 2 3 2 2 4 2

Sample Output 3

8

F - A/AB Insertion

Time Limit: 2 sec / Memory Limit: 1024 MiB

Score : 525 points

Problem Statement

You are given a string S of length N consisting of A and B.

Process a total of Q queries of the following types.

1 i c

Type 1: Change the i -th character of S to c .

2 l r

Type 2:

Let string T be the string obtained by extracting the l -th through r -th characters of the current string S .

If it is possible to obtain string T by the following operation, output Yes; otherwise, output No.

- Starting from the empty string, perform the following two operations any number of times, possibly zero, in any order.
 - Choose any position in the string (possibly the beginning or the end) and insert A there.
 - Choose any position in the string (possibly the beginning or the end) and insert AB there.

Constraints

- N is an integer satisfying $1 \leq N \leq 5 \times 10^5$.
- S is a string of length N consisting of A and B.
- Q is an integer satisfying $1 \leq Q \leq 2 \times 10^5$.
- Each given query is of type 1 or 2.
- Queries of type 1 satisfy the following constraints:
 - i is an integer satisfying $1 \leq i \leq N$, and
 - c is A or B.
- Queries of type 2 satisfy the following constraints:
 - l and r are integers satisfying $1 \leq l \leq r \leq N$.

Input

The input is given from Standard Input in the following format:

```
 $N$   
 $S$   
 $Q$   
Query1  
Query2  
⋮  
Query $Q$ 
```

Here, Query _{i} represents the i -th query, and the input format for a query follows the format described in the problem statement.

Output

Each time a query of type 2 is given, output the answer on one line.

Sample Input 1

```
10  
AABBAABABB  
6  
2 1 10  
1 5 B  
2 1 10  
2 6 8  
1 3 A  
2 1 10
```

Sample Output 1

```
Yes
No
Yes
Yes
```

This input contains six queries.

- Initially, $S = \text{AABBAABABB}$.
- For the 1-st query, $T = \text{AABBAABABB}$, obtained by extracting the 1-st through 10-th characters of S , can be obtained by the following steps, so output Yes.
 - Start from the empty string.
 - Insert AB at the beginning of the empty string, making the string AB.
 - Insert AB right after the 1-st character of AB, making the string AABB.
 - Insert AB at the end of AABB, making the string AABBAB.
 - Insert AB right after the 5-th character of AABBAB, making the string AABBAABB.
 - Insert AB right after the 7-th character of AABBAABB, making the string AABBAABABB.
- For the 2-nd query, change the 5-th character of S to B. As a result, $S = \text{AABBBABABB}$.
- For the 3-rd query, $T = \text{AABBBABABB}$, obtained by extracting the 1-st through 10-th characters of S , cannot be obtained by the operation described in the problem statement, so output No.
- For the 4-th query, $T = \text{ABA}$, obtained by extracting the 6-th through 8-th characters of S , can be obtained by the operation described in the problem statement, so output Yes.
- For the 5-th query, change the 3-rd character of S to A. As a result, $S = \text{AAABBABABB}$.
- For the 6-th query, $T = \text{AAABBABABB}$, obtained by extracting the 1-st through 10-th characters of S , can be obtained by the operation described in the problem statement, so output Yes.

G - Wipeout

Time Limit: 3 sec / Memory Limit: 1024 MiB

Score : 600 points

Problem Statement

There are N cards arranged face down in a row, each with one of the integers $1, 2, \dots, N$ written on its face.

The order of these cards is determined uniformly at random from the $N!$ possible orders.

You know that there is exactly one card with each of the integers $1, 2, \dots, N$ written on its face and that the cards were arranged uniformly at random, but you have no other information about the integers written on the faces of the cards.

You play the following game.

- Initially, let the variable $x = 1$.
- As long as $x \leq N$, repeat the following operation. One operation consists of the following three steps.
 - Specify one card and turn it face up.
 - If the integer written on the card is x , eat that card and add 1 to x .
 - Otherwise, turn that card face down. You can permanently remember the integer written on that card.

You always act so that the expected value of the total number of operations until all cards have been eaten is minimized.

In this case, what is the probability that the total number of operations is K ? Find it modulo 998244353.

► Definition of probability modulo 998244353

Constraints

- All input values are integers.
- $1 \leq N \leq 5 \times 10^5$
- $N \leq K \leq 10^9$

Input

The input is given from Standard Input in the following format:

N K

Output

Output the answer.

Sample Input 1

3 4

Sample Output 1

```
499122177
```

For this input, $N = 3$. Let us call the cards a, b, c in the order they are arranged.

Below is an example of your actions when acting so that the expected value of the total number of operations is minimized.

- First, turn a face up.
- If 1 is written on a , eat that card.
 - Next, turn b face up.
 - If 2 is written on b , eat that card.
 - Next, turn c face up; since 3 is necessarily written on it, eat it. In this case, you eat all cards in three operations, and the probability of this happening is $1/6$.
 - If 3 is written on b , turn that card face down.
 - Next, turn c face up; since 2 is necessarily written on it, eat it. After that, turn b face up and eat it. In this case, you eat all cards in four operations, and the probability of this happening is $1/6$.
- If 2 is written on a , turn that card face down.
 - Next, turn b face up.
 - If 1 is written on b , eat that card.
 - Next, turn a face up and eat it. Next, turn c face up; since 3 is necessarily written on it, eat it. In this case, you eat all cards in four operations, and the probability of this happening is $1/6$.
 - If 3 is written on b , turn that card face down.
 - At this point, it has become clear which integer is written on every one of the cards. So, perform three operations and eat all the cards in order of their numbers. In this case, you eat all cards in five operations, and the probability of this happening is $1/6$.
- If 3 is written on a , turn that card face down.
 - Next, turn b face up.
 - If 1 is written on b , eat that card.
 - Next, turn c face up; since 2 is necessarily written on it, eat it. After that, turn a face up and eat it. In this case, you eat all cards in four operations, and the probability of this happening is $1/6$.
 - If 2 is written on b , turn that card face down.
 - At this point, it has become clear which integer is written on every one of the cards. So, perform three operations and eat all the cards in order of their numbers. In this case, you eat all cards in five operations, and the probability of this happening is $1/6$.

Combining everything, the probability that the total number of operations is 3 is $1/6$, the probability that it is 4 is $1/2$, and the probability that it is 5 is $1/3$.

For this sample, output 499122177, which represents $1/2$ modulo 998244353.

Sample Input 2

```
3 6
```

Sample Output 2

```
0
```

Sample Input 3

```
500000 777777
```

Sample Output 3

```
251612105
```

Hints: AtCoder Beginner Contest 473

Taken from the official editorial, not written by us. Read one hint at a time.

A. Second Half Sum *(editorial, first line)*

The problem can be solved by performing the following two operations: Receive the array A from Standard Input.; Take the sum of the latter half of the array A , and print it.

B. Old Maid *(editorial, first line)*

This problem can be solved by the following procedure: inspect the card one by one, if there is no card with the same integer written on it, confirm that the card is never eaten;; if there is a card with ...

C. Change Schools *(editorial, first line)*

Let M be the maximum number of students in a class before Takahashi transfers.

D. Coefficient Stair *(editorial, first line)*

To exhaustively enumerate objects lexicographically, DFS (Depth-First Search) is effective.

E. K-Divisible Subarrays *(editorial, first line)*

The problem can be rephrased using the mod- K prefix sums S of A , defined as $S_i = \sum_{j=1}^i A_j \pmod K$: Define the score of an integer sequence by the number ...

F. A/AB Insertion *(editorial, first line)*

Interpret A as $+1$, and B as -1 .

G. Wipeout *(editorial, first line)*

It is meaningless to flip the same card three or more times.

Source: <https://atcoder.jp/contests/abc473/editorial>